

EE 210L — Digital Circuits and Logic Design Lab

Lab 4: Decoders and Encoders

Name: _____ Date: _____

Partner(s): _____ Section: _____

Both chips in this lab use active-low outputs. Record the actual measured voltage and then state the logic level. Submit this report on Canvas — see Canvas for the deadline and grading rubric.

Part 1 — Decoder Truth Table

A2	A1	A0	Output that went LOW	Its measured V (V)	A HIGH output's V (V)
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

Part 2 — Enable Behavior

Address held at 011 throughout.

Condition	V at Y3 (V)	Y3 logic level	State of the other outputs
Enabled ($\bar{E}1 = 0$, $\bar{E}2 = 0$, $E3 = 1$)			
$\bar{E}1$ taken HIGH			
E3 taken LOW			

Why does a decoder provide three separate enable pins instead of one?

Part 3 — A Function from Decoder Outputs

$F(A_2,A_1,A_0) = \Sigma m(1, 2, 4, 7)$

Draw the gate arrangement you used to combine the four active-low decoder outputs. Label every pin.

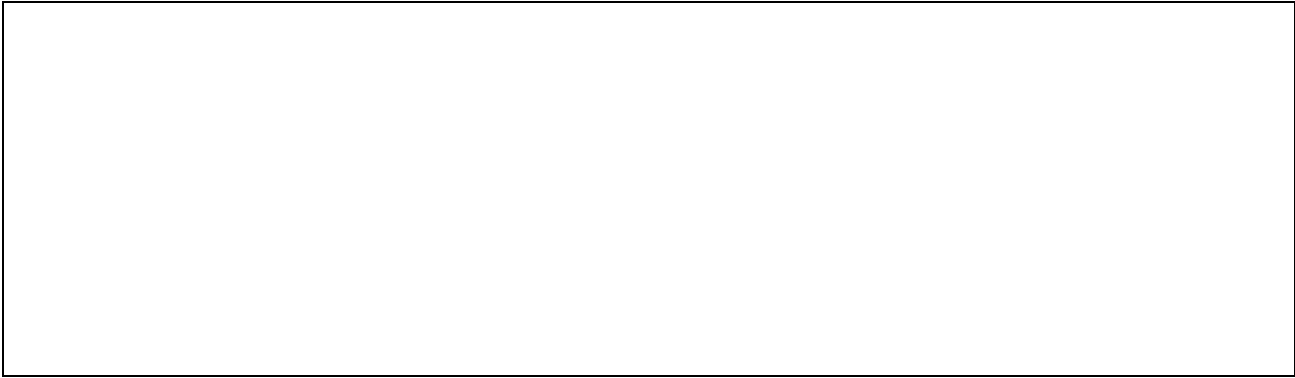


Figure 1: Combining decoder outputs to form F.

A2	A1	A0	Measured V _{out} (V)	Logic Output	Expected F
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

Part 4 — Priority Encoder, One Input at a Time

Input held LOW	$\bar{A}2$	$\bar{A}1$	$\bar{A}0$	Decimal index (after inverting)	$\bar{G}S$	$\bar{E}O$
None (all HIGH)						
$\bar{I}0$						
$\bar{I}1$						
$\bar{I}2$						
$\bar{I}3$						
$\bar{I}4$						
$\bar{I}5$						
$\bar{I}6$						
$\bar{I}7$						

With no input requesting, what distinguishes the output code from a genuine request on input 0?

Part 5 — Priority Resolution

Inputs held LOW	$\bar{A}2$	$\bar{A}1$	$\bar{A}0$	Decimal index reported	$\bar{G}S$
$\bar{I}2$ only					
$\bar{I}2$ and $\bar{I}5$					
$\bar{I}2$, $\bar{I}5$ and $\bar{I}7$					
$\bar{I}2$ and $\bar{I}5$ ($\bar{I}7$ released)					

Which input wins when several request at once, and why is that the behavior an interrupt controller needs?

Discussion and Conclusions

Address at least: why active-low outputs lead to a NAND rather than an OR in Part 3, what $\overline{G}S$ is for, and how $\overline{E}O$ would let you cascade two encoders into a 16-input priority encoder.
